

An object is inside. Why can't we see it?

Learning intention

Explain why an object needs light reaching it and then the eye to be seen.

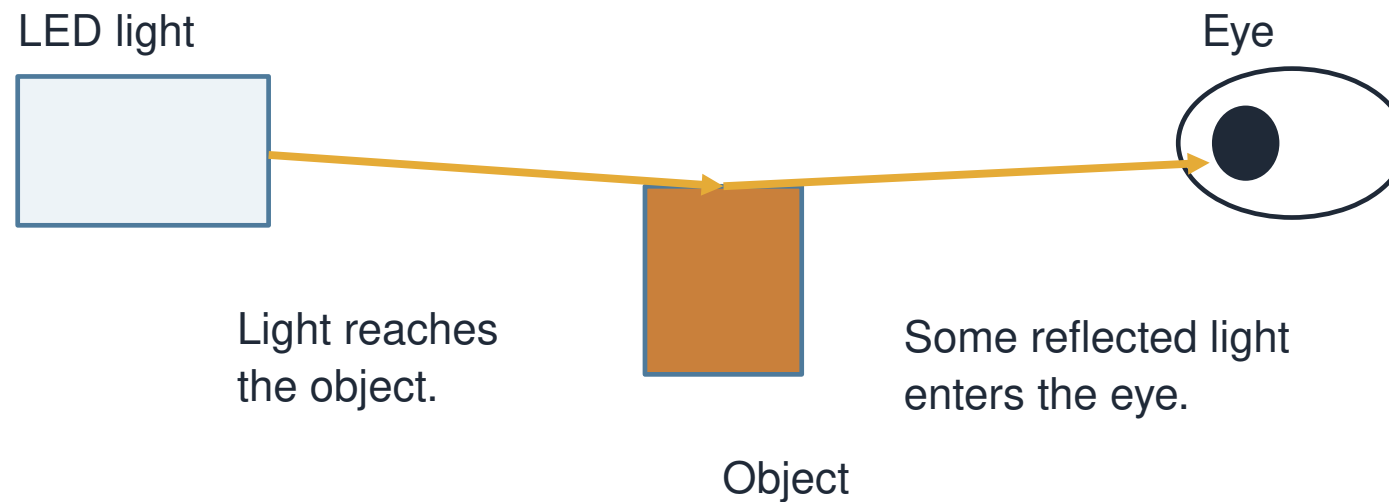
The object has not moved.

Find the missing light path.

What do we already know?

1. Name something that gives out light.
2. Can a plain card make its own light?
3. Which sense uses the eyes?

I follow the light

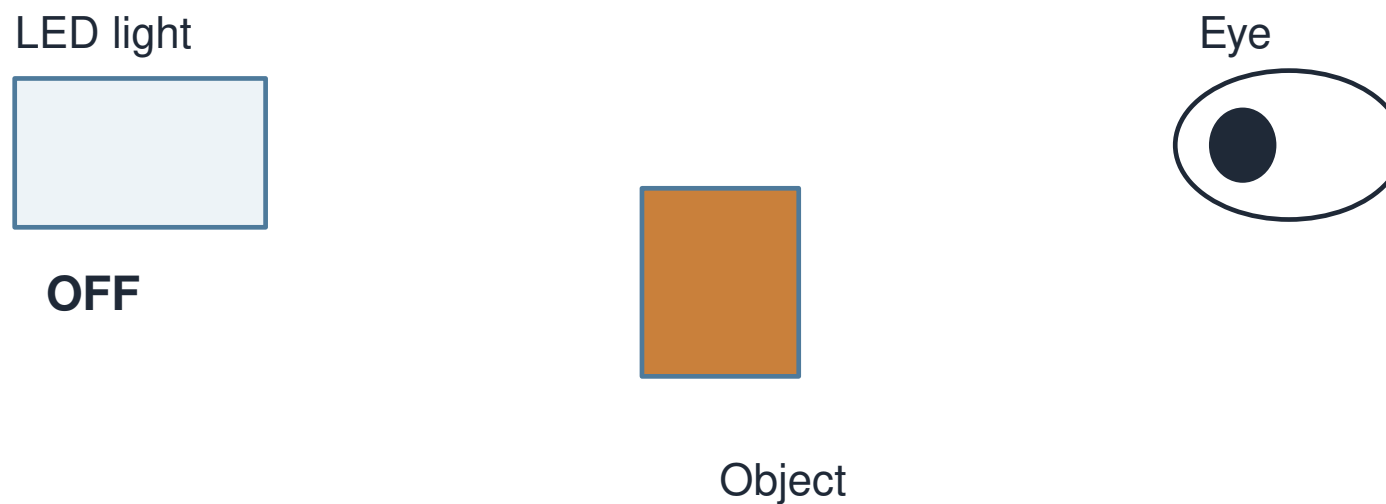


The eye receives light. Arrows show its direction.

The source gives out light. The object reflects some light. Some reaches the eye.

I remove the source

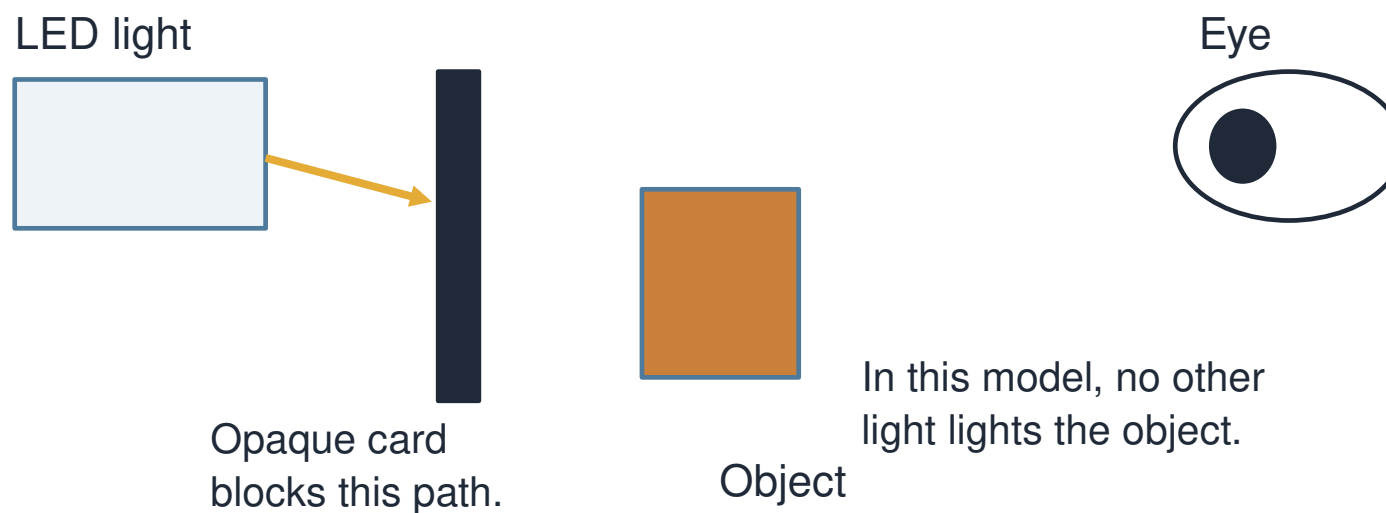
A model of complete darkness



No light reaches the object or the eye.

The object is still present. No light reaches it. Waiting cannot create light.

I block one part of the path



Removing the block can restore the light path.

A card can stop this light. Ask whether another path remains.

The case conference

Read cases A and B. What changed? What stayed in the box?

Case	Box conditions	Constructed observation
A	All light openings sealed; torch off	Object not visible
B	Torch lights object; viewing opening clear	Object visible

Constructed teaching evidence

W1. Which change explains the new view?

Repair the explanation

C: a small light leak; object faintly seen.

D: no light at all; wait one minute.

Which claim needs correcting?

W2. Why are C and D different?

What makes seeing possible?

Choose the best explanation.

- A** Our eyes send light onto the object.
- B** Light from the object enters our eyes.
- C** Waiting makes light appear in total darkness.

Choose your investigation route

Match the cases.

Explain the light path.

Diagnose a leaky box.

Use the evidence

Read or point to one clue.

Record what it supports.

Keep your explanation short.

Make your own science record

Complete your chosen route.

Use a diagram, words or speech.

Check one answer with the evidence.

Check and improve

Does my answer match the evidence?

Have I explained why?

Improve one label or sentence.

Lundy loop: improve our account

Choose how to share one idea.

Listen to another explanation.

Use evidence to improve our account.

What must our explanation include?

Three quick checks

E1. What do we need in order to see?

E2. Does light enter or leave the eye when we see?

E3. Can we see an object with no light at all?